# ECE250: Signals & Systems Quiz-4

**Date:** 17-September-2025 **Total Points:** 10 **Duration:** 15 Minutes

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**Note:**

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

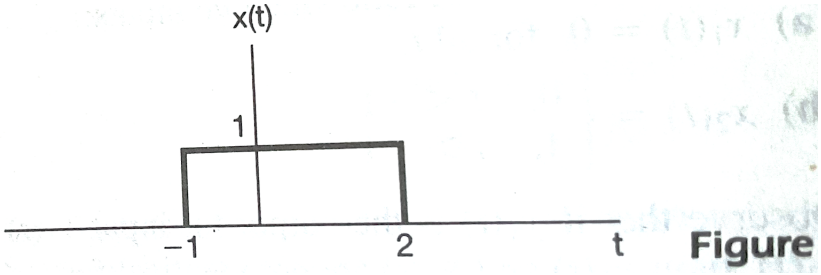
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[CO2] [**Q1**]: Consider an LTI system with input and output related through the equation

$$y(t) = \int_{-\infty}^{t} e^{-(t-\tau)} x(\tau - 3) d\tau$$

(a) What is the impulse response $h(t)$ for this system? **[5 Points]**

(b) Determine the response of the system when the input $x(t)$ is as shown in below Figure. **[5 Points]**



Figure